

LB-AO Source & Evidence Standard

Reference strategy for the Lower-Body Anatomy Ontology
Musculoskeletal Digital Twin Project

Purpose

This document records how sources are selected and what kinds of claims each source may support. The goal is a defensible ontology in which anatomical facts, biomechanical conventions, measured quantities, estimated parameters, and modeling assumptions remain distinguishable.

1. Source Hierarchy

Source class	Primary role	How we use it
OpenStax Anatomy & Physiology 2e	Baseline gross anatomy	Foundational skeletal/joint organization and major landmarks; not a source for detailed muscle-model parameters.
NCBI Bookshelf / StatPearls	Regional and clinical anatomy	Cross-check and expand anatomy, articulations, passive structures, muscle organization, and innervation.
International Society of Biomechanics (ISB)	Biomechanical standardization	Coordinate systems, axis definitions, joint-motion terminology, and reporting conventions.
OpenSim documentation and models	Computational modeling reference	Study how established models translate anatomy into bodies, joints, frames, muscle paths, actuators, and explicit simplifications.
Primary musculoskeletal-model papers	Model derivation	Trace geometry, equations, parameter derivations, and assumptions behind implemented models.
Experimental biomechanics literature	Quantitative evidence	Preferred evidence for numerical properties, with population, method, units, uncertainty, and applicability recorded.

2. OpenStax Anatomy & Physiology 2e

Used for stable gross-anatomical relationships such as the organization of the bony pelvis, hip bones, ilium/ischium/pubis, acetabulum, and major joint structures. It is a useful foundational reference because it describes anatomy without imposing a musculoskeletal simulation architecture.

Evidence role: Use for: anatomical hierarchy, major landmarks, basic articulations. Do not use as the sole justification for specialized quantitative muscle or tissue parameters.

3. NCBI Bookshelf / StatPearls

Used as a detailed regional and clinically oriented anatomy cross-check. Relevant hip and hip-joint chapters help validate articulation, osseous stability, capsule, capsular ligaments, labrum, cartilage, muscle organization, and innervation.

Evidence role: Use for: regional anatomy and corroboration. Important attachment or innervation claims should preferably be corroborated when they become critical to the model.

4. International Society of Biomechanics (ISB)

The ISB Standardization and Terminology Committee recommendations are our preferred starting point for coordinate systems and reporting joint motion. The key hip reference is Wu et al. (2002), Journal of Biomechanics, on joint coordinate systems for

the ankle, hip, and spine.

Evidence role: Use for: coordinate frames, axes, sign conventions, joint-motion definitions, and reporting standards. This prevents an angle such as '+35 degrees hip flexion' from becoming ambiguous.

5. OpenSim

OpenSim is treated as an established computational modeling reference, not as the authority on anatomical truth. Models such as Gait2392/Gait2354 demonstrate how anatomy is reduced to rigid bodies, joints, coordinates, musculotendon paths, actuators, and model-specific assumptions.

Evidence role: Use for: implementation patterns, comparison models, abstraction choices, and tracing parameter/model lineage. OpenSim-specific choices remain modeling choices.

6. Primary Model Literature

Published papers underlying established musculoskeletal models are preferred over software documentation when justifying how geometry, equations, or parameters were derived. The Delp-model lineage is particularly relevant to lower-extremity musculoskeletal modeling.

Evidence role: Use for: model equations, muscle geometry, moment-arm methodology, parameter derivation, and documented assumptions. Whenever possible, trace important model values further to the underlying experimental evidence.

7. Experimental Biomechanics Literature

Peer-reviewed experimental studies become the highest-priority evidence for quantitative properties. A number without context is not treated as a universal anatomical constant.

Evidence role: Record with a quantitative value: unit, measurement/estimation method, source population, coordinate convention where relevant, scaling procedure, uncertainty, model applicability, and source citation.

8. Evidence Must Support Specific Claims

A citation attached to an entity does not automatically validate every field in that entity. Evidence records should state exactly which claims the source supports.

```
evidence:
- source_id: ISB_WU_2002
supports:
- coordinate_system
- joint_motion_definition

- source_id: STATPEARLS_HIP
supports:
- articulation
- ligament_relationships

- source_id: DELP_MODEL_PAPER
supports:
- musculotendon_geometry
- model_parameter_derivation

- source_id: OPENSIM_GAIT2392
supports:
- computational_representation
- model_comparison
```

9. Claim Classification Rule

Before information enters LB-AO, classify the claim. This determines what evidence is appropriate and prevents modeling assumptions from being mistaken for anatomy.

Claim type	Preferred evidence
Anatomical structure / relationship	Established anatomical references, preferably corroborated for important details.
Coordinate / motion definition	ISB or another explicitly documented biomechanical standard.
Biomechanical behavior	Peer-reviewed experimental or biomechanical literature.
Numerical parameter	Original experimental literature where feasible, including method/population/unit metadata.
Model implementation	OpenSim and peer-reviewed musculoskeletal-model literature.
Our simplification	Explicit MODEL_ASSUMPTION record; never presented as anatomical evidence.

10. Quantitative Parameter Standard

The upcoming quantitative-property schema will preserve provenance rather than storing context-free constants.

```
maximum_isometric_force:
value: ...
unit: N
claim_type: estimated_parameter
source_id: ...
measurement_basis: ...
population: ...
scaling_method: ...
uncertainty: ...
applicability: ...
coordinate_convention: ...
notes: ...
```

11. Core References

Reference	Role
OpenStax, Anatomy & Physiology 2e - Pelvic Girdle and Pelvis / Selected Synovial Joints	Foundational gross anatomy.
NCBI Bookshelf / StatPearls - Anatomy, Bony Pelvis and Lower Limb, Hip; Hip Joint	Regional hip anatomy and clinical-anatomical detail.
Wu et al. (2002), Journal of Biomechanics 35(4):543-548, DOI 10.1016/S0021-9290(01)00222-6	ISB recommendations for ankle, hip, and spine joint-coordinate systems.
OpenSim - Gait2392/Gait2354 documentation and Musculoskeletal Models documentation	Established computational-model structure, frames, joints, musculotendon actuators, and documented assumptions.
Delp et al. and subsequent lower-extremity musculoskeletal-model literature	Primary literature supporting model geometry, parameter lineage, and computational biomechanics.

12. Working Standard

Anatomical fact != biomechanical convention != measured quantity != estimated parameter != model assumption.

LB-AO will preserve those distinctions, attach evidence to individual claims, and keep source/model context with quantitative values. This is the standard to apply as the ontology expands through the lower body.